

MCA 2002 – EXAMINATION PAPER

1. The number of divisors of 2002 excluding 1 and 2002 is
 1) 10 2) 12
 3) 14 4) 16
2. The sum of the digits in the sum of the two-digit prime numbers ending with 1 is
 1) 6 2) 7
 3) 8 4) 9
3. The poles stand vertically on the ground. If the difference between the distance between their tops and the distance between their feet is half the difference between the heights of the poles, then the ratio of the distance between the tops of the poles to the distance between the feet of the poles is
 1) 5 : 4 2) 5 : 3
 3) 4 : 3 4) 3 : 2
4. The price of an article is decreased by 20% and then increased by 20%. Effectively, the price of the article is
 1) not changed
 2) increased by 4 and $\frac{1}{6}\%$
 3) increased by 2%
 4) decreased by 4%
5. A man walks 1 metre due east, then 2 metres due north, then 3 metres due west and then 4 metres due south. Then he walks 5 metres due east and so on, walking cyclically as in the first round by increasing 1 metre in each direction and completes his walking, finally by walking 100 metres due south. At the end, the distance of the man from the starting point is nearly
 1) 35.35 metres 2) 70.70 metres
 3) 141.40 metres 4) 5050 metres
6. There are 4 vessels containing milk mixed with water. The ratios of the milk to water in the 4 vessels are 3 : 4, 4 : 3, 5 : 2 and 6 : 1. The amount of mixture in the 4 vessels are 1 litre, 2 litres, 3 litres and 4 litres respectively. If the contents of the 4 vessels are poured into another big vessel of capacity more than 10 litres, the ratio of the milk to water in the new vessel is
 1) 4 : 3 2) 5 : 2
 3) 9 : 5 4) 11 : 3
7. In a class room, there are certain number of benches and certain number of students. If 4 students sit in a bench, one bench is completely empty. If 3 students sit in a bench, one student has no place to sit in a bench. The number of students in the class room is
 1) 16 2) 28
 3) 40 4) 64
8. A and B are two single digit numbers. If $A73B92$ is divisible by 99, then
 1) $A = 9$ and $B = 6$ 2) $A = 6$ and $B = 9$
 3) $A = 5$ and $B = 1$ 4) $A = 4$ and $B = 2$
9. By working at the rate of 6 hrs./day, a person agreed to complete a work in 160 days. However, in order to complete the work in advance, he worked at the rate of 8 hrs./day, 10 hrs./day and 12 hrs./day for the second, third and fourth quarter of the work respectively. In how many days in advance he completed the work?
 1) 36 2) 40
 3) 46 4) 50

1 1 2 * 3 2 4 4 5 2 6 2 7 1 8 2 9 3

10. Three couples are arranged in a row for a photograph. The couples should not be separated and no man should be adjacent to the wife of another man. The number of possible arrangements is

- 1) 720 2) 48
3) 24 4) 12

11. Swapna's salary is 120% of the salary of her brother Sam whose salary is Rs. 2,000 more than 90% of the salary of his brother Santosh. If the total salary of the three persons is Rs. 49,100 then Swapna's salary exceeds the salary of Santosh by

- 1) Rs. 3,600 2) Rs. 3,100
3) Rs. 2,400 4) Rs. 500

12. If the radius of a circle is increased by 25%, the area of the circle is increased by

- 1) 25% 2) 50%
3) 56.25% 4) 625%

13. To see a circus the ticket rate is Rs. 25 for gents, Rs. 20 for ladies and Rs. 10 for children between 3 and 12 years. On a particular day, the total number of tickets sold is 2140 and the total collection is Rs. 35,800. Which one of the following figures is infeasible to be the number of ladies who saw the circus on that day?

- 1) 660 2) 720
3) 805 4) 825

14. From a square board the maximum possible circle is cut out and the remaining area is called A. Another identical square board is first divided into 64 equal squares as in chess board and the maximum possible circle is cut out from each one of the 64 squares. The remaining area is called B. Which one of the following is true?

- 1) $A = 4B$ 2) $B = 4A$
3) $A = 2B$ 4) $A = B$

15. In the duration of one hour between 5 p.m. and 6 p.m., at what time, the hour hand and the minute hand coincide?

- 1) 5 hrs. 27 and 3/11 min.
2) 5 hrs. 29 and 1/11 min.
3) 5 hrs. 25 min.
4) 5 hrs. 30 min.

16. A and B together can complete a work in 10 hrs. B and C together can complete the work in 20 hrs. C and A together can complete that work in 12 hrs. If A, B and C join together, in how many days that work can be completed?

- 1) 5 and 5/7 hrs. 2) 7 hrs.
3) 8 hrs. 4) 8 and 4/7 hrs.

Directions for Questions 17 and 18 :

A farmer on being asked how many animals he had in his farm, replied, they are all buffaloes except 48, all cows except 47, all goats except 46 and all horses except 45 and no animals other than these.

17. How many animals did the farmer have?

- 1) 186 2) 124
3) 93 4) 62

18. How many cows are there in the farm?

- 1) 36 2) 16
3) 15 4) 14

19. The time taken by an express train to reach Chennai from Trichy with the usual speed is 5 hrs. 30 min. If the speed is increased by 6 km. hr., the time is reduced by 30 min. If the speed is decreased by 6 km. hr., then the time taken is increased by

- 1) 30 min. 2) 36 min.
3) 36 min. 40 sec. 4) 35 min. 20 sec.

10 ____ 4 11 ____ 1 12 ____ 3 13 ____ * 14 ____ 4 15 ____ 1 16 ____ 4 17 ____ 4 18 ____ 3 19 ____ 1

20. Which of the following statements are true?

- A) If a perfect square is divided by 3, the remainder is always 0 or 1
 B) If a perfect square is divided by 7, the remainder can never be 3 or 5
 C) If a perfect cube is divided by 5, the remainder can never be 2 or 3
 D) If a perfect cube is divided by 9, the remainder is always 1 or 8

- 1) A and B only 2) A and C only
 3) B and D only 4) C and D only

21. In a competitive examination, there are 100 multiple choice type of questions. For answering a question correctly 1 mark will be given. For answering a question wrongly $\frac{1}{3}$ mark will be deducted. However, there is no penalty for not attempting a question. If at least one-third of the questions attempted by a student are answered wrongly, what is the maximum possible mark that he/she can score?

- 1) 58 and $\frac{2}{3}$ 2) 66 and $\frac{2}{3}$
 3) 46 and $\frac{2}{3}$ 4) 55

22. The first prime year in the 21st century is

- 1) 2003 2) 2009
 3) 2011 4) 2017

23. A man has to swim to reach B from A along a straight path and has to swim back to A in 30 minutes. C is the mid point of AB. By swimming at a speed of p m/sec., from A to C and then at a speed of r m/sec. from C to B, he reached B from A in 20 minutes. Then by swimming at a speed of q m/sec.; if he reached A exactly in time, then which of the following are in harmonic progression?

- 1) p, q, r 2) $p, 2q, r$
 3) $p, 4q, r$ 4) $2p, q, 2r$

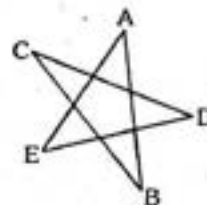
24. Which of the following three measures cannot be sides of a triangle?

- 1) 2 cm, 3 cm, 4 cm
 2) 3 cm, 4 cm, 5 cm
 3) 4 cm, 8 cm, 16 cm
 4) 7 cm, 11 cm, 15 cm

25. A merchant purchased 100 kg. of sugar for Rs. 1,400. He sold at the rate of Rs. 14.63 per kg. However he weighed only 950 gms. instead of 1 kg. What is the actual profit for the merchant?

- 1) 10% 2) 9%
 3) 5% 4) 4.5%

26. What is the sum of the 5 angles A, B, C, D and E in the following type of five cornered star?



- 1) 180 degrees 2) 240 degrees
 3) 300 degrees 4) 360 degrees

27. There are 100 students in a class and they were given two tests. In the first test, only 37 passed whereas in the second test 49 passed. At least how many students failed in both the tests?

- 1) 12 2) 14
 3) 43 4) 51

28. A land lord wanted to give some area of land for his sincere most worker. He called that worker and gave a fence of 120 metres long and told him to cover as much area as possible by fencing the land in a rectangular form. If the worker is clever enough, how much area of land he will get?

- 1) 900 sq. m. 2) 3600 sq. m.
 3) 2000 sq. m. 4) 800 sq. m.

20 1 21 2 22 2 23 2 24 3 25 2 26 4 27 2 28 1

29. There are 1248 students in college to choose a President. If 4 candidates contest for this post, what should be the minimum number of votes to be polled in favour of the successful candidate with absolute majority?

- 1) 313 2) 417
3) 624 4) 625

30. If 150% of a number is 0.24, then what will be 0.24% of it?

- 1) 150 2) 0.0384
3) 0.000384 4) 0.000864

31. How many times 7 appear when the integers between 10 and 100 are written?

- 1) 17 2) 18
3) 19 4) 20

32. If the cost of a cone ice cream is Rs. 12, what is the cost of the lower half portion of the ice cream in the cone having the vertex?

- 1) Rs. 1.50 2) Rs. 3.00
3) Rs. 4.50 4) Rs. 6.00

33. A train 250 metres long running at a speed of 60 km/hr crosses a 350 metres long train coming in the opposite direction in 16 seconds. What is the speed of the other train?

- 1) 37.5 km/hr 2) 60 km/hr
3) 72 km/hr 4) 75 km/hr

34. If the price of cooking gas is increased by 5%, by how much percentage a house-wife must reduce the consumption of cooking gas so as not to increase or decrease the expenditure on cooking gas?

- 1) 5 and 5/19%
2) 4 and 17/21%
3) 4 and 16/21%
4) 5%

35. If $f(x)$ denotes the greatest integer less than or equal to the cube root of $3x$, then the sum of the series $\sum f(n)$ where n varies from 1 to 10 is

- 1) 13 2) 19
3) 20 4) 39

36. If Rs. 10,000 is deposited for a period of one year at the rate of 8% per annum and if the interest is accumulated to the capital every 6 months, what will be interest for one year?

- 1) Rs. 416 2) Rs. 816
3) Rs. 832 4) Rs. 1.632

37. In the year 1996, the age of a mother was three times the age of her only son. In the year 2006, the age of the mother will be only two times the age of her son. In the year 2002, the ratio of the age of the mother to the age of her son is

- 1) 12 : 5 2) 9 : 4
3) 8 : 3 4) 5 : 2

38. There are two outlet taps, A and B in a tank. When both taps are opened, the tank with full of water is drained in 45 minutes. If the tap A is opened and the tap B is closed, the tank will be drained in 1 hour. If the tap B is opened and the tap A is closed, how long will it take to drain the tank?

- 1) 2 hrs. 2) 3 hrs.
3) 2 hrs. 30 min. 4) 4 hrs.

Directions for Questions 39 and 40 :

A basket contains certain number of apples. Anita takes 1 less than 50% of the apples. Then Banu takes 1 less than 50% of the left out. Then Chitra takes 1 less than 50% of the balance. Finally only 3 apples are left out in the basket.

29 * 30 3 31 2 32 4 33 4 34 3 35 3 36 2 37 2 38 2

39. How many apples were there in the beginning in the basket?

- 1) 10 2) 16
3) 18 4) 20

40. How many apples that Banu has taken?

- 1) 4 2) 3
3) 2 4) 1

41. The next term in the sequence 1, 4, 10, 20, 35, 56, 84, 120, is

- 1) 165 2) 175
3) 180 4) 192

42. Among the four groups of letters, three are alike in some way while one is different. Find the one which is different

- 1) CSUF 2) LBDO
3) PGJS 4) UMOX

43. If SHIP is written as PEFM, then BOAT may written as

- 1) DQCW 2) ERDX
3) YLXQ 4) ZMYR

44. Find the odd man out

- 1) TESRIS 2) HATFER
3) HOTREM 4) LOVINI

45. Insert the missing number

243 (250) 368

317 () 438

- 1) 242 2) 323
3) 324 4) 342

46. In the following list, one of the words is my pass word. If I were to tell you any letter of my pass word, then you would be able to tell me the number of vowels in my pass word. Which is my pass word?

- 1) CASH
3) TIME

- 2) SNOW
4) IDEA

Directions for Questions 47 - 50 :

In a group of seven girls, Aruna is taller than Divya but shorter than Hema who is not the tallest girl. Divya is not as short as Priya who is not the shortest. Shilpa is shorter than 4 girls including Geetha and taller than the other girls. No data is available about the remaining girl Bama.

47. The tallest girl is

- 1) Divya 2) Aruna
3) Bama 4) Geetha

48. The shortest girl is

- 1) Bama 2) Divya
3) Shilpa 4) Hema

49. The girl who stands middle in the height is

- 1) Divya 2) Aruna
3) Priya 4) Hema

50. The number of girls taller than Aruna is

- 1) One 2) Two
3) Three 4) Four

51. Find the missing number in the sequence 2, 7, 10, 63, ?, 215, 50

- 1) 26 2) 28
3) 80 4) 126

Directions for Questions 52 - 55 :

Three adjacent faces of a cube having a common vertex are painted with green colour and the other three faces are painted with red colour. The cube is cut into 216 equal small cubes.

39 1 40 3 41 1 42 4 43 3 44 * 45 3 46 4 47 4 48 1
49 1 50 2 51 1

52. How many small cubes have green paint on one face and red paint on another face and no paint on the remaining four faces?

- 1) 12 2) 24
3) 48 4) 96

53. How many small cubes have no paint on any of the faces?

- 1) 48 2) 64
3) 96 4) 125

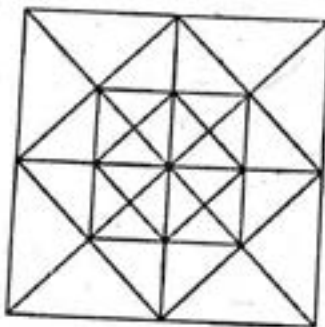
54. How many small cubes have paint only on one face?

- 1) 96 2) 72
3) 64 4) 48

55. How many small cubes have red colour on two faces and green colour on one face and no paint on the other three faces?

- 1) 2 2) 3
3) 4 4) 6

For Questions 56 and 57 :



56. In the above figure, how many squares are there?

- 1) 16 2) 20
3) 24 4) 28

57. In the above figure, how many triangles are there?

- 1) 72 2) 84
3) 96 4) 100

58. Find the missing figure :

3, 7, 17, 39, 85, ?

- 1) 179 2) 189
3) 147 4) 255

59. Find the missing pair of letters :

IR LO ? RI UF XC

- 1) OL 2) ON
3) NO 4) RO

60. Which of the figures 1, 2, 3, 4 given in the answers will logically fit into the next figure in the following sequence?



- 1)  2) 
3)  4) 

61. The first mechanical computer designed by Charles Babbage was called

- 1) Abacus 2) Analytical Engine
3) Calculator 4) Processor

62. If 4A9C in hexadecimal system is x in octal system then x is

- 1) 43254 2) 45234
3) 11247 4) 74211

63. The 10's complement of the octal number 2002 in decimal equivalent is

- 1) 7998 2) 7997
3) 8973 4) 8974

52 2 53 2 54 1 55 1 56 2 57 2 58 1 59 1 60 3 61 2
62 2 63 *

64. The simplified form of the Boolean expression $(A' + B' + C') (A' + B + C') (A' + B + C)$ is
 1) $A + B'C$ 2) $A' + BC'$
 3) $A + BC'$ 4) A'
65. The parity bit is added for the purpose of
 1) coding
 2) indexing
 3) error-detection
 4) controlling
66. Which of the following is the most appropriate unit for measuring the storage capacity of a hard disk?
 1) byte 2) megabyte
 3) bit 4) terabyte
67. A temporary storage area, attached to the CPU, for I/O operations is a
 1) buffer 2) channel
 3) core 4) register
68. Which of the following hardware components is the most volatile?
 1) ROM 2) RAM
 3) PROM 4) EPROM
69. Which of the following memories is an optical memory?
 1) Floppy Disk
 2) Core Memories
 3) Bubble Memories
 4) CD-ROM
70. The only things moving around inside a computer are
 1) bytes 2) electrons
 3) 1's and 0's 4) words
71. What is the name of the network topology in which there are bi-directional links between each possible node?
 1) Ring 2) Star
 3) Tree 4) Mesh
72. Which of the following might prevent a program from being modified in the future?
 1) logic errors
 2) lack of program documentation
 3) syntax errors
 4) pseudocode
73. What is the primary requisite of a good computer programmer?
 1) Mathematical mind
 2) Artistic mind
 3) Logical mind
 4) Scientific knowledge
74. A robot's "arm" is also known as
 1) actuator 2) end effector
 3) manipulator 4) servo mechanism
75. DNS refers to
 1) Data Number Sequence
 2) Domain Name System
 3) Disk Numbering System
 4) Digital Network Service
76. Find the odd man out
 1) FTP 2) POP
 3) SAP 4) TCP
77. Which of the following is ascending order of a data hierarchy?
 1) bit-byte-record-field-file-database
 2) byte-bit-field-record-file-database
 3) bit-byte-field-record-file-database
 4) byte-bit-record-file-field-database

64 2 65 3 66 2 67 4 68 2 69 4 70 3 71 1 72 * 73 3
 74 * 75 4 76 2 77 3

78. The binary number representing the hexadecimal number ACE is

- 1) 1010110011110
- 2) 0000000000001
- 3) 111011001010
- 4) 1010000000000

79. A portion of a BASIC program is shown below

```
10 DIM A(3, 3)
20 MAT A = CON
```

Execution of this program will cause the matrix A to be defined as

- 1) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- 2) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$
- 3) $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- 4) $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

80. Suppose I contains 6 and J contains 12.

Find the final value of I after the following FORTRAN program segment

```
IF (J. GT. I) GO TO 10
```

```
I = I + 1
```

```
GO TO 20
```

```
10 I = J
```

```
20 I = I + J
```

- 1) 18
- 2) 19
- 3) 24
- 4) 37

81. An electric motor

- 1) generates mechanical energy
- 2) generates electrical energy
- 3) converts mechanical energy into electrical energy
- 4) converts electrical energy into mechanical energy

82. What is the critical temperature of water?

- 1) 100° C
- 2) 0° C
- 3) -273° C
- 4) 374° C

83. Who is considered to be the father of medicine?

- 1) Hippocrates
- 2) Salk
- 3) Jenner
- 4) Redi

84. Water is a fluid medium of relatively low

- 1) viscosity
- 2) surface tension
- 3) density
- 4) specific heat

85. Blood pressure is measured by an instrument called

- 1) Anemometer
- 2) Barometer
- 3) Sphygmomanometer
- 4) Hygrometer

86. The value of acceleration due to gravity is minimum

- 1) at the poles
- 2) at the centre of the earth
- 3) at the equator
- 4) it is same everywhere

87. The stars are not visible in the day time because

- 1) they are much smaller in comparison to the sun
- 2) they are at a very great distance from the earth
- 3) of sun's brightness during day time
- 4) they appear only during the night time

88. The most liquid of all assets is

- 1) money
- 2) shares
- 3) bonds
- 4) fixed deposits

78 1 79 2 80 3 81 4 82 1 83 1 84 3 85 1 86 1 87 3
88 1

89. If the time at Chennai is 6.00 a.m. then the time at that moment at Singapore is approximately

- 1) 3.30 a.m. 2) 8.30 a.m.
3) 2.30 a.m. 4) 9.30 a.m.

90. Ornithology is

- 1) the study of plants
2) the study of birds
3) the study of bones
4) the study of inscriptions

91. 8th March is celebrated as

- 1) Mother's day
2) Women's day
3) Parent's day
4) Children's day

For Questions 92 - 94 :

Each of the question below consists of a word in capital letters followed by four words. Choose the one that is most nearly similar in meaning to the word in capital letters.

92. THESPIAN

- 1) foreigner
2) day dreamer
3) magician
4) actor

93. REGAL

- 1) royal 2) major
3) basic 4) entertaining

94. FORMIDABLE

- 1) venerable
2) dangerous
3) adjustable
4) impenetrable

For Questions 95 - 97 :

Each questions below consists of a word printed in capital letters followed by four words. Choose the word that is most nearly opposite in meaning to the word in capital letters.

95. LUCID

- 1) obscure 2) arrogant
3) ornate 4) embroiled

96. ERRATIC

- 1) unromantic 2) free
3) popular 4) steady

97. HYPOTHETICAL

- 1) rational 2) fantastic
3) opposed 4) axiomatic

For Questions 98 - 100 :

Select the pair of words which are related in the same way as the pair of words in capital letters are related to each other.

98. EGG : YOLK

- 1) melon : rind 2) cell : nucleus
3) spider : web 4) corn : stalk

99. INAUGURATE : PRESIDENT

- 1) campaign : candidate
2) promote : student
3) succeed : governor
4) install : officer

100. GLOSSARY : WORDS

- 1) atlas : maps
2) catalogue : dates
3) thesaurus : rhymes
4) lexicon : numbers

89 2 90 2 91 2 92 4 93 1 94 1 95 1 96 4 97 4 98 2
99 2 100 1